

Bhasha / Language	Hinglish (Hindi + English)
Tools	VS Code + Python 3.10+
Duration	2 hours per day
Batch	Beginners + Intermediate (mixed)
Workshop	Smart Traffic System — 18 Days

Contents

Day 1	Variables & Data Types if/elif/else match-case
Day 2	Lists Dictionaries for loop while loop

Teacher ke liye note

Yeh PDF screen pe open karke padhao — students VS Code mein code likhein.
Har topic mein: pehle real-life analogy, phir code, phir ek chhota exercise.
Code blocks exactly waise type karein jaise diye gaye hain — output bhi check karein.

Day 1

Python Fundamentals

Variables & Data Types | if/elif/else | match-case

Topic 1 — Variables aur Data Types

Yeh kya hota hai?

Variable = ek dabba (box) jisme hum data rakhte hain.

Jaise ghar mein alag-alag dabbe hote hain — chawal ka dabba, dal ka dabba — Python mein bhi alag-alag cheezein alag variables mein store karte hain.

Traffic system se example:

Hamara traffic system in cheezein track karta hai:

lane_name = 'North' — kaun si lane hai (text = str)

vehicle_count = 12 — kitni gaadiyaan hain (pura number = int)

avg_speed = 45.5 — average speed km/h (decimal = float)

is_emergency = False — ambulance hai ya nahi (haan/na = bool)

Code — VS Code mein type karo:

```
# Variables banao

lane_name      = 'North'

vehicle_count  = 12

avg_speed      = 45.5

is_emergency   = False


# Print karo

print('Lane:', lane_name)

print('Vehicles:', vehicle_count)

print('Speed:', avg_speed, 'km/h')

print('Emergency:', is_emergency)
```

Output:

Lane: North
Vehicles: 12
Speed: 45.5 km/h
Emergency: False

Try it yourself:

Apni pasand ki lane ke liye variables banao.

— lane_name ko 'South' karo

— vehicle_count ko 25 karo

— is_emergency ko True karo

Print karke dekho output kya aata hai.

Topic 2 — if / elif / else**Real life analogy**

Traffic light pe yahi hota hai:

Agar (if) gaadiyaan zyada hain → green zyada time do

Warna agar (elif) thodi hain → normal time do

Warna (else) → minimum time do

Syntax:

```
vehicle_count = 18

if vehicle_count > 20:

    print('Bahut zyada traffic – 60s green')

elif vehicle_count > 10:

    print('Thoda traffic      – 30s green')

else:

    print('Kam traffic        – 15s green')
```

Output:

Thoda traffic – 30s green

Comparison operators — yaad rakhne wale:

>	zyada hai	count > 10
<	kam hai	count < 5
>=	zyada ya equal	count >= 20
<=	kam ya equal	count <= 0
==	bilkul equal	lane == 'North'
!=	equal nahi	state != 'GREEN'

Try it yourself:

Yeh program likhne ki koshish karo:

vehicle_count = 5 set karo

if/elif/else se check karo kitna green time milega

Phir count 25 karo — output change hoga?

Topic 3 — match-case (Python 3.10+)

Kab use karte hain?

Jab ek variable ki exact value ke hisaab se alag kaam karna ho.

Traffic mein: signal ka current state check karna — 'RED', 'GREEN', ya 'YELLOW'.

if/elif se better tab hota hai jab options fixed aur clear hon.

Syntax:

```
signal_state = 'GREEN'

match signal_state:

    case 'GREEN':

        print('Gaadiyaan chal sakti hain!')

    case 'YELLOW':

        print('Slow down karo, rukne wale hain.')

    case 'RED':

        print('Ruko — signal red hai.')

    case _:
```

```
print('Unknown signal state!')
```

Output:

Gaadiyaan chal sakti hain!

if/elif vs match-case — kab kya?

if/elif → Jab condition range-based ho (count > 20, speed < 10)

match → Jab exact fixed values check karni hon ('RED', 'GREEN', 'YELLOW')

Try it yourself:

match-case se yeh banao:

lane_direction = 'North' set karo

case 'North' / 'South' / 'East' / 'West' ke liye alag print karo

case _ mein 'Invalid lane' print karo

Mini Project: Traffic Signal Checker

Ab teen topics ko ek saath use karte hain — variables, if/elif/else, aur match-case:

```
# Day 1 Mini Project — Traffic Signal Checker
```

```
lane_name      = 'North'
```

```
vehicle_count  = 18
```

```
signal_state   = 'RED'
```

```
# if/elif — green time decide karo
```

```
if vehicle_count > 20:
```

```
    green_time = 60
```

```
elif vehicle_count > 10:
```

```
    green_time = 30
```

```
else:
```

```
    green_time = 15
```

```
print(lane_name, 'lane -', vehicle_count, 'vehicles')

print('Recommended green time:', green_time, 'seconds')

# match-case — current state batao

match signal_state:

    case 'GREEN':

        print('Signal abhi GREEN hai — gaadiyaan ja sakti hain')

    case 'RED':

        print('Signal RED hai — rukna padega')

    case 'YELLOW':

        print('Signal YELLOW — slow down karo')

    case _:

        print('Signal state unknown')
```

Output:

```
North lane — 18 vehicles
Recommended green time: 30 seconds
Signal RED hai — rukna padega
```

Ghar pe challenge (optional)

vehicle_count ko 25 karo — green_time change hoga?
signal_state ko 'GREEN' karo — kya print hoga?
Ek naya variable is_emergency = True add karo.
Agar is_emergency True hai toh green_time = 90 forcefully set karo.

Day 2

Collections aur Loops

Lists | Dictionaries | for loop | while loop

Day 1 ka Quick Recap

Variable = data store karne ka dabba (int, float, str, bool)

if / elif / else = condition ke hisaab se alag kaam karna

match-case = fixed values ke liye clean switching

Topic 1 — List

Real life analogy

List = ek line mein khadi gaadiyaan.

Jaise North lane mein gaadiyaan ek ke baad ek khadi hain — pehle number 0, phir 1, phir 2...

Hum list mein se koi bhi gaadi nikal sakte hain, nayi add kar sakte hain, ya sab dekh sakte hain.

List banao aur use karo:

```
# North lane ki gaadiyaan
```

```
north_lane = ['Car', 'Truck', 'Bike', 'Car', 'Bus']
```

```
print(north_lane)           # Poori list
```

```
print(north_lane[0])        # Pehli gaadi → Car
```

```
print(north_lane[-1])       # Aakhri gaadi → Bus
```

```
print(len(north_lane))      # Kitni gaadiyaan → 5
```

```
# Nayi gaadi aayi
```

```
north_lane.append('Bike')
```

```
print('After append:', north_lane)
```

```
# Pehli gaadi nikal gayi (signal green tha)
```

```
north_lane.pop(0)

print('After pop:', north_lane)

# Gaadi hai ki nahi?

print('Truck hai?', 'Truck' in north_lane)
```

Output:

```
['Car', 'Truck', 'Bike', 'Car', 'Bus']
Car
Bus
5
After append: ['Car', 'Truck', 'Bike', 'Car', 'Bus', 'Bike']
After pop:    ['Truck', 'Bike', 'Car', 'Bus', 'Bike']
Truck hai? True
```

Try it yourself:

Apni ek list banao south_lane ke liye 4 gaadiyaan daalo.

Print karo: pehli gaadi, aakhri gaadi, aur total count.

Ek aur gaadi append karo aur dobara count print karo.

Topic 2 — Dictionary

Real life analogy

Dictionary = har lane ka apna labelled counter.

Jaise ek register mein likhte hain: North = 12, South = 5, East = 20...

Dict mein data key:value pairs mein hota hai — key lane ka naam, value count.

Dictionary banao aur use karo:

```
# Har lane ka vehicle count

lane_counts = {

    'North': 12,

    'South': 5,

    'East' : 20,

    'West' : 8
```

```
}

# Values access karo

print('North mein:', lane_counts['North'])    # 12

print('East mein:', lane_counts['East'])      # 20


# Value update karo (camera ne naya count diya)

lane_counts['North'] = 15

print('Updated North:', lane_counts['North'])


# Nayi lane add karo

lane_counts['Flyover'] = 3


# Saari keys aur values

print('Keys:', list(lane_counts.keys()))

print('Values:', list(lane_counts.values()))


# Sabse zyada traffic wali lane

busiest = max(lane_counts, key=lane_counts.get)

print('Busiest lane:', busiest)
```

Output:

```
North mein: 12
East mein: 20
Updated North: 15
Keys:  ['North', 'South', 'East', 'West', 'Flyover']
Values: [15, 5, 20, 8, 3]
Busiest lane: East
```

Try it yourself:

Apna dictionary banao 4 lanes ke saath.

South lane ka count 25 karo (update).

max() use karke busiest lane find karo — kya change hua?

Topic 3 — for Loop

Real life analogy

for loop = ek ek karke sabko check karna.

Jaise CCTV camera sabhi 4 lanes ko ek ek karke scan karta hai.

Jab tak lanes hain tab tak scanning hoti rehti hai — automatically.

for loop — list ke saath:

```
lanes = ['North', 'South', 'East', 'West']
```

```
for lane in lanes:
```

```
    print('Checking lane:', lane)
```

Output:

```
Checking lane: North
```

```
Checking lane: South
```

```
Checking lane: East
```

```
Checking lane: West
```

for loop — dictionary ke saath:

```
lane_counts = {'North': 12, 'South': 5, 'East': 20, 'West': 8}
```

```
for lane, count in lane_counts.items():
```

```
    if count > 10:
```

```
        green_time = 30
```

```
    else:
```

```
        green_time = 15
```

```
    print(lane, ': ', count, 'vehicles ->', green_time, 'sec green')
```

Output:

```
North : 12 vehicles -> 30 sec green
South : 5 vehicles -> 15 sec green
East  : 20 vehicles -> 30 sec green
West  : 8 vehicles -> 15 sec green
```

range() ke saath for loop:

5 se 0 tak countdown — signal switch hone se pehle

```
for i in range(5, 0, -1):

    print(i, '...')

print('Signal change!')
```

Output:

```
5 ...
4 ...
3 ...
2 ...
1 ...
Signal change!
```

Try it yourself:

lane_counts dictionary use karo.
for loop se har lane print karo.
Saath mein ek message bhi print karo:
count > 15 → 'Heavy traffic'
count > 8 → 'Moderate traffic'
else → 'Light traffic'

Topic 4 — while Loop**Real life analogy**

while loop = jab tak koi condition sach hai tab tak kaam karta raho.
Traffic light cycle sochlo: jab tak system on hai, signals chalte rehte hain.
Camera bhi while loop mein hi kaam karta hai — jab tak video band na ho, frames aate rehte hain.

while loop — traffic light cycle:

```
import time

cycle_count = 0

while cycle_count < 3:    # Sirf 3 cycles chalao

    print('--- Cycle', cycle_count + 1, '---')

    print('GREEN - 5 seconds')

    time.sleep(1)          # Workshop mein 1 sec use karo

    print('YELLOW - 2 seconds')

    time.sleep(0.5)

    print('RED - 5 seconds')

    time.sleep(1)

    cycle_count = cycle_count + 1

print('3 cycles complete!')
```

Output:

```
--- Cycle 1 ---
GREEN - 5 seconds
YELLOW - 2 seconds
RED - 5 seconds
--- Cycle 2 ---
... (repeats) ...
3 cycles complete!
```

for vs while — kab kya?

for loop → Jab pata ho kitni baar chalana hai (4 lanes, 10 frames, range(5))

while loop → Jab condition pe depend kare (jab tak emergency nahi, system chalo)

Try it yourself:

while loop use karke yeh banao:

vehicle_count = 0 se start karo

Jab tak count < 10, print karo 'Waiting... count = X'

Har baar count += 1 karo

Loop ke baad print karo 'Lane full!'

Mini Project: Automatic Signal Simulator

Ab sab cheez ek saath — list, dict, for loop, while loop, aur conditions:

```
# Day 2 Mini Project — Automatic Signal Simulator
```

```
import time
```

```
# Har lane ka vehicle count (baad mein camera se aayega!)
```

```
lane_counts = {
```

```
    'North': 12,
```

```
    'South': 5,
```

```
    'East' : 20,
```

```
    'West' : 8
```

```
}
```

```
total_cycles = 2
```

```
cycle = 0
```

```
while cycle < total_cycles:
```

```
    print(f'\n==== Cycle {cycle + 1} ====')
```

```
# Sabse busy lane dhundo
```

```
    busiest = max(lane_counts, key=lane_counts.get)
```

```

# Har lane ki status print karo

for lane, count in lane_counts.items():

    if lane == busiest:

        status = 'GREEN'

        green_time = 10 + count * 2

    else:

        status = 'RED  '

        green_time = 0

    print(f' {lane:6}: {count:3} vehicles | {status} | {green_time}s')

    time.sleep(1)    # Real system mein actual green_time hoga

    cycle += 1

print('\nSimulation complete!')

```

Output:

```

===== Cycle 1 =====
North : 12 vehicles | RED    | 0s
South :  5 vehicles | RED    | 0s
East  : 20 vehicles | GREEN  | 50s
West  :  8 vehicles | RED    | 0s
===== Cycle 2 =====
... (repeats) ...
Simulation complete!

```

Yeh kahan kaam aayega?

Yahi logic Day 14-15 mein real cameras se connect hoga.
lane_counts values tab YOLO detection se automatically aayengi.
Aaj hum manually set kar rahe hain — logic same rahega!